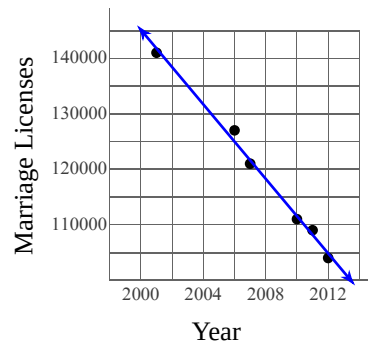


- 3) The number of marriage licenses issued by Clark County Nevada, the county where Las Vegas is located, has been decreasing since the year 2000:

Year	Marriage Licenses
2001	141,000
2006	127,000
2007	121,000
2010	111,000
2011	109,000
2012	104,000

This can be modeled by the equation $y = -3358.1x + 6861400$ where x is the year and y is the number of marriage licenses issued.

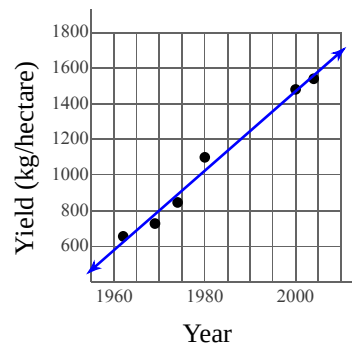


- a) According to the model, how many marriage licenses were issued in 2004? Round your answer to the nearest hundred.
- b) Using this model, how many marriages licenses would you expect to be issued in 2023? Round your answer to the nearest hundred.
- c) According to the model, in what year did Clark County issue 140,000 marriage licenses? Disregard years before 1990. Round your answer to the nearest year.

- 4) With the help of scientists, farmers in Cameroon have been able to produce more and more grain per hectare each year. Here are the crop yields for several years:

Year	Yield (kg/hectare)
1962	656
1969	728
1974	846
1980	1,100
2000	1,480
2004	1,540

The crop yield can be described by the equation $y = 22.3x - 43100$ where x is the year and y is the grain yield in kilograms per hectare (kg/ha).



- a) According to the model, what was the crop yield in 1988? Round your answer to the nearest whole number.
- b) Assuming that this trend continues, what crop yield is predicted for the year 2022 by the model? Round your answer to the nearest whole number.
- c) The model indicates that a crop yield of 1300 kg/hectare was achieved in what year? Round your answer to the nearest year.